GPT: Generative Pre-trained Transformer

↓ generates text

↓ parts trained separately

↓ text transformers speech, video, image
↳ predict the next word

Parts:

Embedding → Attention → FFNN
↓
Unembedding

Tokens: (input)

- words: a, the, nice
- word parts: nice/est
- punctuation marks: , .

Embedding:

token → vector (meaning)
includes position !!!

Attention:

context (what are neighbour words)
e.g.: Queen Elisabeth or Queen pawn

Multi layer perception

This is where the real knowledge of the model is

Unembedding

vector → token (with propabilities)

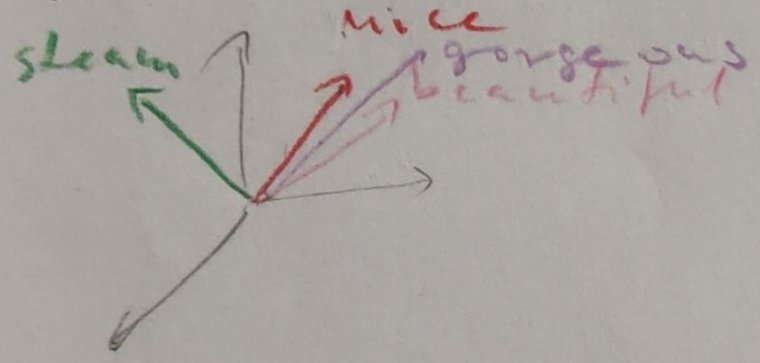
Embedding: (GPT3)

tokens ~ 50.000

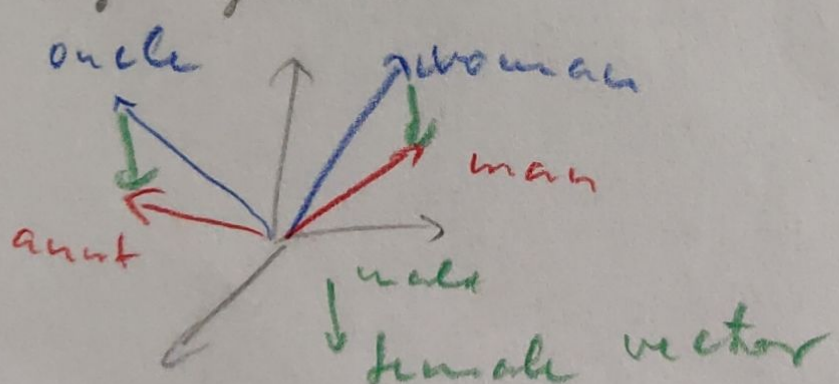
dimension of the embedding space 12.288
(matrix $6,2 \cdot 10^8$ values)

one hot encoded token to a 12k dim vec.

- words with similar meaning have similar vectors



- some directions have abstract meaning



eg: plural
(mice - mouse) + two >
>(- -) + three
etc.

Problems: e.g. queen ≠ king + female vector

queen: - sovereign
- chess piece
- band

Context is important: it modifies, or specifies the meaning of a word

Context size (GPT-2048)

↳ how much text is considered at a time

Unembeddings: after transformation and generation we will have a 12,288 dimension vector about the meaning of the next word.

unembedding matrix + softmax =

= probabilities for tokens.

Size: same as embedding (50,000 × 12288)

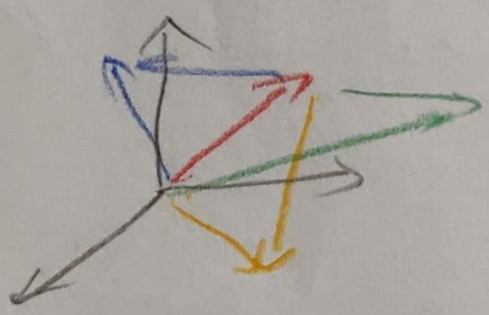
Transformers: "Attention is all you need" (2017)

idea: modify embedded vector to include the context: e.g. Queen Elizabeth

Queen singer

Queen pawn opening

ex:



Harry Potter's favourite teacher was Lupin

Lupin should come from the word was!

Harry Potter in front of teacher should bring the vector around Hogwarts and favourite should select Lupin.

Words will ask questions and others answer it. If the answer positive then modify the meaning.

Example:

Question: Are you an adjective?
- this should be high for nouns

Answer (key) I am! only for adjective

Modify: the value of the asking word by a product of a value matrix and the adjective

Matrices: Q, K, V

Attention: $\text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$

	Embedding	query / key	$\vec{E}_1 \cdot W_Q = \vec{Q}_1$	$\vec{E}_2 \cdot W_Q = \vec{Q}_2$	$\vec{Q}_3$	$\vec{Q}_4$ ← dot products
Hanny	$\vec{E}_1$	$\cdot W_K = \vec{K}_1$				0,1
Potter's	$\vec{E}_2$	$\cdot W_K = \vec{K}_2$	0			0,1
favourite	$\vec{E}_3$	$\cdot W_K = \vec{K}_3$	0	0		0,8
teacher	$\vec{E}_4$	$\cdot W_K = \vec{K}_4$	0	0	0	0,-

← causality

Key/Query dimension (GPT 128) | not zero for translation

So both W_Q and W_K

The value matrix transforms an embedding vector to an embedding space, so it should be 12288 × 12288. But it is too large. low rank transformation

$\underline{\underline{V}} = \underline{\underline{V}}_{\downarrow} \cdot \underline{\underline{V}}_{\uparrow}$ $\quad \underline{\underline{V}}_{\downarrow} : 12288 \times 128, \underline{\underline{V}}_{\uparrow} : 128 \times 12288$

One attention head: $K, Q, V_{\downarrow}, V_{\uparrow} \doteq 4 \cdot 1,5 \cdot 10^6$ par.

= Multi-headed attention

queries run in parallel with diffent matrices (GPT 96) ($6 \cdot 10^9$ par)

Add all changes to the embedding to the word

- Multi layer perceptron (FFNN)

These two are repeated many times (GPT 96 layers of these)

2/3 of the weights are in the perceptron layers which hold the knowledge